Subspace Transfer for Low-Resource Bantu MT — Probe-Guided Rule-Based Concord Rewriting for Kitaita

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Abstract

1 Introduction

Kitaita (also Kidawida or Dawida; ISO 639-3 `dav`) is a Bantu language spoken by an estimated 200,000 people (Eberhard et al., 2024) in the Taita Hills of Taita-Taveta County, in southeastern Kenya (Sakamoto, 2003). Within Guthrie's referential classification of the Bantu languages, as updated by Maho (2009), it is placed at E74, in the Kilimanjaro-Taita group – which also contains the Chaga languages of Mount Kilimanjaro – specifically within the closely related Taita-Sagalla subgroup, rather than in the Sabaki group to which Swahili (Kiswahili, Guthrie G42) belongs (Nurse and Hinnebusch, 1993). Swahili and Kitaita are therefore both Bantu languages with a long history of geographic proximity and contact (the Taita Hills lie along historical trade routes inland from the Swahili coast), but they are not close relatives within the Bantu family tree: their noun-class and agreement systems are structurally similar in kind, as is typical across Bantu, but differ in the specific concord forms and class assignments this project has to establish empirically rather than assume. This distinction matters directly for natural language processing: Swahili is one of roughly 200 languages covered by NLLB (NLLB Team et al., 2022) and has existing dependency treebanks, embeddings, and other digital resources, while Kitaita has no entry in NLLB, no dependency treebank, and only a small, very recent amount of digitized text – a community-based effort to build the first NLP corpus for Kidaw'ida only completed in 2025 (Mbogho et al., 2025). For practical NLP purposes, Kitaita remains essentially unrepresented.

Like other Bantu languages, both Swahili and Kitaita organize their nouns into a system of *noun classes*: every noun belongs to one of around fifteen to twenty class pairs (singular paired with a corresponding plural class), each historically marked by its own prefix, following the numbering convention established by Meinhof and still used today (classes are conventionally numbered 1 through ∼23, with odd numbers singular and the following even number their plural counterpart) (Katamba, 2003). A noun's class is not just a lexical label: it drives *concord*, or grammatical agreement, across the rest of the sentence – adjectives, possessives, demonstratives, and the verb's own subject and object markers all take a prefix that echoes the controlling noun's class (Katamba, 2003). For example, in Swahili, class 1 *m-* (*mtu*, "person") requires the matching subject marker *a-* on its verb (*a-na-keti*, "he/she is seating"), while class 7 *ki-* (*kitabu*, "book") requires *ki-* (*ki-na-keti*, in a different sentence, "it is seating") – the same grammatical relation, realized with a different, noun-class-dependent prefix. This is the mechanism this project calls *concord rewriting*: correctly reinflecting these agreement-bearing prefixes for Kitaita's own class assignments, which, as discussed below, do not always coincide with Swahili's.

Bantu verbs are also highly agglutinative: rather than a single inflectional suffix as in English, a Bantu verb is built by stacking a sequence of prefixal and suffixal morphemes around the root, each encoding one grammatical category, following a canonical template reconstructed for Proto-Bantu by Meeussen (1967) and still broadly valid across the family today (Nurse and Devos, 2019): subject marker – negation – tense/aspect/ mood (TAM) – object marker – ROOT – (derivational extensions, e.g. causative, applicative, passive) – final vowel. The Swahili example above, *a-na-keti* (sm-PRS-seat, "he/she is seating"), already shows three of these slots filled; a fuller form like *a-li-ki-ketia* (sm-PST-om-seat, "he/she sat on it") fills four. Both the concord prefixes and this verbal template are exactly what this project's pipeline has to identify (which noun controls which agreement slot) and reinflect (substituting the correct, independently-verified Kitaita morpheme at each slot) – not a fixed string substitution, since the same grammatical relation surfaces as a different prefix depending on the controlling noun's class in both languages.

Kitaita represents one of hundreds of Bantu languages which are not represented in NLP. In this work we try to exploit its relatedness with Kiswahili for the purpose of machine translation and the techniques developed might be applicable to other Bantu languages.

2 Related Work

2.1 Exploiting language relatedness for MT

Three recent precedents attempt to exploit a related high-resource language for a low-resource target, each at a different level of the pipeline and with a different operating assumption about how closely the two languages must resemble each other.

Zoph et al. (2016) pretrain an NMT *parent* model on a high-resource pair (French-English, 300M English tokens) and initialize a *child* model with its weights before fine-tuning on the low-resource pair's own parallel data; this reduces, but does not eliminate, the amount of child-language parallel data required – their smallest case, Urdu, still used 200K English-side tokens (∼11.4K sentences). Notably, relatedness between parent and child is not required for the method to work at all: French transfers positively even to unrelated Hausa, Turkish, and Uzbek, though a closely-matched synthetic parent gives a larger gain (+6.7 vs. +4.3 BLEU). This method does not transfer directly to Kitaita: no Kitaita-side parallel corpus of any kind existed at a usable scale until Mbogho et al. (2025)'s 2025 crowd-sourcing effort, and that corpus pairs Kidaw'ida with Swahili rather than English. Even setting data availability aside, a neural model fine-tuned on tens of thousands of sentences would need many examples per (noun class, TAM, relation) combination to learn Kitaita's concord system reliably – exactly the "low-count event" weakness Zoph et al. (2016) themselves cite as NMT's core failure mode, and exactly what this project's rule-based concord tables are built to avoid by trusting a single native-speaker-verified judgment ($n = 1$) where a gradient-trained model could not.

Pourdamghani and Knight (2019) take a different approach: a character-level noisy-channel "cipher" model, trained via EM on monolingual data alone, converts a related higher-resource language's parallel text into synthetic low-resource/English pairs, which then augment a conventional phrase-based MT system. Their method is explicitly built for language pairs that are mostly cognates with the same word order, differing mainly in orthography (e.g. Afrikaans/Dutch, Serbian/Bosnian); it has no model of syntax or morphological agreement, since none of their six tested pairs require one. Swahili and Kitaita do not satisfy this precondition – they diverge in concord morphology, not just orthography – so a character-substitution cipher would likely fail the way Pourdamghani and Knight's own weakest case did (Malaysian/Indonesian, where the languages were so similar the method "cannot learn much more than copying"), just from the opposite direction of divergence.

Sakajo et al. (2025) instead transfer cross-lingual *vocabulary* directly: exploiting a property of BPE tokenizers (removing a subword forces a fallback to its component pieces), bilingual dictionaries are used to estimate target-language subword embeddings and adapt a pretrained model's vocabulary without retraining on parallel or even monolingual target-language text, outperforming prior vocabulary-transfer baselines (random initialization, FOCUS, WECHSEL) when dictionary coverage is adequate. The authors explicitly flag morphologically complex languages and incomplete dictionary coverage as limiting cases – both apply directly to Kitaita, and together describe precisely the naive dictionary-substitution baseline this project's null hypothesis has to beat: vocabulary transfer alone, with no explicit treatment of agreement morphology.

2.2 Probing methodology

This project's structural-identification stage (§3.2) builds directly on the probing-classifier literature. Hewitt and Manning (2019) introduce the *structural probe*: a single linear transformation of a pretrained model's contextual word representations, fit so that squared Euclidean distance under the transformation approximates tree distance in the sentence's dependency parse. That such a transformation exists at all, for representations never given parse trees as supervision, is the foundational evidence that syntax is linearly decodable from contextual embeddings – the premise this project's own probes rest on.

A probe achieving high accuracy, however, does not by itself show that the representation encodes the property being probed, since a sufficiently expressive probe can learn to approximate any labeling function, including an arbitrary one, given enough parameters and training data. Hewitt and Liang (2019) formalize this concern with *control tasks*: a probe is trained to predict a deliberately randomized relabeling of the same inputs (e.g. a fixed but arbitrary part-of-speech tag per word type, independent of context), and *selectivity* – the gap between real-task accuracy and control-task accuracy – is what actually distinguishes a probe that reads structure already present in the representation from one that has simply memorized the task. This project adopts that discipline as a standing rule: every probe result reported here is accompanied by a random-relabel control and its selectivity, not the raw accuracy alone.

Chi et al. (2020) extend the structural probe across languages, showing that a probe trained on one language's treebank can recover syntactic structure for a *different* language never seen during probe training, provided both languages share a multilingual encoder's representation space – grammatical relations are not just linearly decodable, but decodable in a way that is largely language-agnostic within such a space. This is the direct precedent for this project's own finding that a relation probe trained only on Swahili data transfers zero-shot to English (0.90/0.88 Swahili vs. 0.90/0.88 English nsubj/obj accuracy) via NLLB's shared encoder, without ever training on English-labeled data.

Finally, Müller-Eberstein et al. (2022) ("DepProbe") generalize structural probing from unlabeled tree *distance* to full *labeled, directed* dependency trees, recovered by a lightweight probe rather than a full parser. This project's own relation probe is architecturally descended from this move: a single multiclass probe, trained on gold Universal Dependencies-style annotation (SWH_UDT and AfriSUD), predicts a 21-way label set covering every agreement-bearing relation this pipeline needs (`nsubj`, `obj`, `amod`, `nmod:poss`, `acl:relcl`, among others) directly from NLLB encoder representations, resolved per sentence via a cross-candidate win condition rather than a fixed classification threshold.

Beyond these four, this project trained and validated several further probes specific to its own agreement-identification needs, and reports negative results alongside the positive ones, consistent with its standing rule that a null hypothesis must be beaten, not assumed. A dedicated part-of-speech

probe (NOUN/VERB/ADJ/PRON/OTHER) and a binary *edge-presence* probe (does a given verb have a subject at all, as distinct from *which* noun is the subject) were added after the relative, cross-candidate win condition used elsewhere was found structurally unable to express "no candidate should win" for a verb with no real subject argument; the edge-presence probe validated with real selectivity but still failed the single real case (a Swahili participial *-ki-* construction) it was built to resolve, and was not put into production (§3.2). On the negative side: raw attention (both encoder self-attention and encoder-decoder cross-attention) was tested twice, independently, as a training-free alternative to a trained probe for subject/object tracking; both attempts looked convincing on short, hand-built sentences and both failed an adversarial, position-confound-controlled check on real corpus data, revealing the signal to be a linear-position artifact rather than grammatical tracking – the one attention-based signal that survived the same check targets a different relation entirely (noun→adjective concord). Separately, a cross-lingual check in the style of Chi et al. (2020) – train a probe on English's control-construction annotation (`nsubj:xsubj`) and test it zero-shot on Swahili participial (*-ki-*) verb subjects, which SWH_UDT itself marks with the same relation only 5 times in 226 occurrences – has so far been scoped only as far as data availability (English UD treebanks carry enough `xsubj` edges; German, Spanish, French, and Finnish treebanks carry none at all) and has not actually been trained or run (§3.2).

3 Methodology

3.1 Pipeline overview

The overall pipeline consists of four stages:

1. **Generate Swahili.** Translate the source sentence into Swahili with NLLB – a standard use of NLLB, since Swahili is one of its 200 covered languages – which also supplies the encoder context the later stages probe.

2. **Identify structure.** A probe recovers each token's role and the dependency relations between tokens. For example, *Mtoto anaogopa nyoka mkubwa* ("the child is afraid of the big snake") is first broken into part-of-speech tags – *mtoto* (child, noun), *anaogopa* (is afraid of, verb), *nyoka* (snake, noun), *mkubwa* (big, adjective) – and then into the relations between them: *anaogopa* takes *mtoto* as subject and *nyoka* as object, and *mkubwa* modifies *nyoka*.

3. **Resolve Kitaita class agreement for nouns.** Each noun's Swahili agreement class is looked up, and the dictionary supplies the corresponding Kitaita class – which need not match:

   - *mtoto*: Swahili animate class 1 (agreement *a-*) → Taita class 1 (agreement *u-*);
   - *nyoka*: Swahili animate class 1 (agreement *a-*, via the animacy override) → Taita class 9 (agreement *i-*).

4. **Substitute, reinflect, and reassemble.** Vocabulary is substituted from the dictionary, and every agreement-bearing slot identified above is reinflected for its controlling noun's Kitaita class:

   - noun: *mtoto* → *mwana*;
   - subject marker: *a-* (Swahili class 1 singular) → *wa-* (Taita class 1 marker *u-*, surface-shifted to *wa-* before the present-tense marker);
   - tense marker: *na-* (Swahili present) → *-we-* (Taita present);

- verb root: *ogopa* → *obua*;
- noun: *nyoka* → *choka*;
- adjective: *mkubwa* (class 1 "big") → *mbaa* (class 9 "big").

Reassembled: *mtoto anaogopa nyoka mkubwa* → *mwana waweobua choka mbaa*.

Bantu class numbers ("class 1," "class 9") are used above purely for exposition. Kitaita's actual agreement groups, as established empirically rather than assumed (§3.3), are tracked by their own concord markers (e.g. `u`, `i_nasal`, `ghu`, `chi`) and do not map one-to-one onto Swahili's numbered classes.

3.2 Structural identification

All probes in this section are linear read-outs trained on frozen NLLB encoder representations, layer 12 throughout (a systematic layer sweep remains open – see below), with hyperparameters selected on a validation split disjoint from the reported test split and the project's standard optimizer settings (learning rate 10^{-4}, gradient norm clipped to 1.0). Two different control designs are used below, and we keep them distinct rather than folding both into one blanket "control task" claim: the labelled-edge and classification probes (relation, concord-source, part-of-speech, edge-presence, governor-attachment) are each checked against a Hewitt and Liang (2019) *random-relabel* control – the identical architecture trained on a fixed-but-arbitrary relabelling of the same inputs – and we report *selectivity* (real accuracy/F1 minus control accuracy/F1) throughout; the *structural (distance) probe*, by contrast, has so far only been checked against an *untrained, randomly-initialized* probe baseline, a weaker check that does not rule out an architecture simply capable of memorizing an arbitrary labelling.

Structural (distance) probe. Following Hewitt and Manning (2019), a single linear transform is fit so that squared Euclidean distance under the transform approximates unlabelled dependency-tree distance. A first run underperformed an untrained random probe on all three languages tested – a clear bug signal, traced to NLLB's raw encoder hidden states (layer-12 norms ∼1766–12,700) being far larger than the norm range (∼20–25) the probe's default learning rate was calibrated for; rescaling every token embedding to a fixed norm of 20 before training fixed this (English in-sample jumped to +0.922 Spearman correlation between predicted and gold tree distance, UUAS 0.816, matching or beating Hewitt and Manning's own published numbers). With that fix, an English-trained probe (373 UD EWT sentences, zero target-language data) transfers zero-shot to real, human-annotated gold trees in three further languages: Spanish +0.546/0.419 (random baseline 0.181/0.307), Finnish +0.495/0.479 (random 0.252/0.326), and Swahili ($n = 175$ gold SWH_UDT sentences) +0.587/0.499 (random 0.264/0.354) – Swahili transfer is at least as strong as both the related and the typologically distant comparison language. On the same held-out Swahili test split ($n = 35$), the English-transfer probe (+0.599/0.492) slightly *outperforms* one trained directly on the available 140 Swahili sentences (+0.568/0.464), attributed to 140 sentences being too few to train a good probe from scratch; the production pipeline therefore uses the English-trained probe and reserves the Swahili gold set for evaluation. A full layer sweep and a sentence-alone-vs.-paragraph context-stability check – both named in the project's own staged plan – remain open for this probe specifically (the equivalent context-stability check for encoder *attention*, a separate, non-probe signal, was run and found no effect, but that result does not transfer to this probe's own training). The resulting tree is also word-level only: NLLB's subword tokenization only coincidentally overlaps with true morpheme boundaries, so this probe identifies *which* noun controls *which* verb and via which

relation type, but not which part of the verb's surface form to rewrite – that is the agreement-target and reinflection stages' job (§3.3 and below).

Labelled/classification probes. Table 1 reports every remaining trained probe against its random-relabel control.

Table 1: Labelled/classification probes: target feature, held-out real vs. random-relabel control performance, and selectivity (layer 12 throughout).

Probe	Target feature	Train n	Real	Control	Selectivity
Relation (21-way)	labelled UD relation per head–dependent pair (`nsubj`, `obj`, `amod`, `det`, `case`, ...)	140/35 sent. (701 pairs)	acc 0.825 / F1 0.694	acc 0.138 / F1 0.050	+0.686 / +0.644
Concord source: case (*wa*)	binary: is this noun the concord source for an associative/case slot	108/27 sent.	acc 0.922 / F1 0.847	acc 0.539 / F1 0.457	+0.384 / +0.391
Concord source: det/amod	binary: concord source for a determiner/adjective-modifier slot	1376/195[a]	acc 0.965 / F1 0.934	acc 0.541 / F1 0.487	+0.425 / +0.447
Concord source: `acl:relcl`	binary: concord source for a relative-clause slot	AfriSUD[a]	acc 0.944 / F1 0.891	acc 0.533 / F1 0.561	+0.411 / +0.330
Concord source: `nmod:poss`	binary: concord source for a possessive slot	AfriSUD[a]	acc 0.983 / F1 0.966	acc 0.533 / F1 0.222	+0.450 / +0.743
Part-of-speech	UPOS: NOUN/VERB/ADJ/PRON/other	6645 test tok.[b]	acc 0.906	acc 0.695	+0.211
Edge-presence	binary: does this verb have a real subject/object arc at all	SWH_-UDT+AfriSUD	–[c]	–[c]	+0.430[c]
Governor attachment	for a non-finite (*-ki-*) verb, which token is its real syntactic head	1713/1757[d] sent. (80/20)	acc 0.931 / F1 0.704	acc 0.527	+0.459

[a]AfriSUD-scale training; the first SWH_UDT-only pass (175 sentences) was too small to validate `acl:relcl` ($n = 13$ positive) or `nmod:poss` ($n = 1$) at all, and is superseded here. [b]SWH_UDT+AfriSUD combined, 80/20 split. [c]Reported directly as held-out selectivity rather than a real/control accuracy pair; a separate targeted multi-verb discrimination check gives accuracy 0.851 (95% CI [0.824, 0.875], $n = 624/733$). [d]After a data- contamination fix (15.7% of *-ki-* detections had been mislabelled nouns); selectivity before the fix was +0.404.

Deployment-condition generalization. The classification numbers above are on a fixed, balanced test split; the four concord-source probes are also checked under the pipeline's actual deployment condition, a cross-candidate win rate against every other noun in the sentence, not just the gold pair: case/*wa* 0.845 [0.731, 0.916] ($n = 58$); det/amod 0.923 [0.877, 0.953] ($n = 195$); `acl:relcl` 0.836 [0.717, 0.911] ($n = 55$); `nmod:poss` 1.000 [0.796, 1.000] ($n = 15$).

Cross-lingual zero-shot transfer (relation probe). The same relation probe, trained only on 140 Swahili sentences and never retrained, transfers zero-shot to English UD EWT with combined accuracy 0.905 [0.862, 0.935] for subjects and 0.877 [0.831, 0.912] for objects (chance ≈ 0.048 for a 21-way argmax). A further adversarial, position-confound stratification of the same (verb, `nsubj`,

obj) triples shows the pattern that rules out a simple linear-position explanation: on Swahili, subject identification scores *higher* on the position-confounded split (adversarial $n = 51$, acc 0.941) than the easy one (non-adversarial $n = 12$, acc 0.917; combined $n = 63$, acc 0.937 [0.848, 0.975]); object identification shows the easy-split number (non-adversarial $n = 61$, acc 0.902) but its adversarial cell ($n = 2$) falls below this project's own $n \geq 8$ floor and is not claimed (combined $n = 63$, acc 0.889 [0.788, 0.945]). The identical stratification replicates on the zero-shot English evaluation, where both cells clear the $n \geq 8$ floor: subject non-adversarial $n = 180$/acc 0.900, adversarial $n = 72$/acc 0.917 (combined $n = 252$, acc 0.905); object non-adversarial $n = 126$/acc 0.841, adversarial $n = 126$/acc 0.913 (combined $n = 252$, acc 0.877) – the adversarial-beats-or-matches-easy pattern holding independently in a second language is read as evidence the probe tracks grammatical relations rather than word order, though we note EWT and SWH_UDT share a register (news text), so some of this transfer may be genre-assisted rather than purely language-general.

Caveats. The part-of-speech probe's headline numbers hold, but a targeted diagnostic ($n = 7$, reported qualitatively per this project's own $n < 8$ floor) found it is not actually tracking finiteness or nominal use as intended – it tracks whether an infinitive governs an object, which happens to correlate with the VERB label – so it cannot resolve the participial (*-ki-*) subject problem it was partly built for. The edge-presence probe (built for that same gap, where only 5 of 226 *-ki-* verbs in SWH_UDT have a gold `nsubj` edge at all) validated with real selectivity but still got the single motivating case wrong and was never put into production; a follow-up diagnosis found its score tracks clause-local string proximity rather than true gold attachment. Its successor, the governor-attachment probe – repurposing the same architecture to find a non-finite verb's real syntactic head rather than ask whether a subject edge exists – validated with higher selectivity but only a modest, not statistically significant improvement over the heuristic it replaces (win rate 0.548 vs. 0.435, $n = 62$, paired $p \approx 0.08$); a compositional pipeline built around this probe, not a single classifier, is the current production path for *-ki-* verbs. Finally, the English-`xsubj`-to-Swahili zero-shot check referenced in §2 has been scoped for data availability only (English UD treebanks carry enough `xsubj` edges; German, Spanish, French, and Finnish treebanks carry none at all) and has not actually been trained or run.

3.3 Kitaita agreement classes

Kitaita's agreement classes were established empirically, noun by noun, rather than assumed from Swahili's. This project tracks twelve Kitaita classes, each identified by its own subject-marker/concord prefix: `u`, `wi`, `ghu`, `ghi`, `ji`, `i`, `i_nasal`, `chi`, `ku`, `lu`, `ri`, and `va`. Table 2 cross-tabulates these twelve Kitaita classes against the Swahili noun classes of this project's own 319-noun Swahili agreement lexicon (`swahili_noun_lexicon_agreement_v2.py`, derived from NLLB concord behaviour, numbered following Katamba (2003)), for every noun where a Kitaita translation and class are both resolvable: 229 of the 319 nouns (72%), of which 60 translations are direct native-speaker judgements and 169 are the project's own corpus-mined statistical candidates (Dice-coefficient co-occurrence against the parallel Swahili–Kitaita Bible corpus). Swahili's class 1/2 (animate singular/plural) is reported as ANIMATE-SG/ ANIMATE-PL rather than by number, because the source lexicon itself already folds in Swahili's animacy override (an inherently animate noun such as *nyoka*, "snake," takes class-1 agreement regardless of its historical shape-class membership) rather than keeping the two separate.

Two patterns stand out, and both motivate this project's standing refusal to assume Swahili structure transfers unchanged. First, correspondence is one-to-many, not one-to-one, and unevenly so: Kitaita `wi` aligns perfectly with Swahili ANIMATE-PL (19/19, 100%), but Kitaita `i` – the single

Table 2: Kitaita class (rows) by Swahili class (columns): noun counts, $n = 229$. Row/column labels follow this project's own notation (Table 1 and §3.2 use the same relation labels; see above for why ANIMATE-SG/PL replace Swahili class numbers 1/2 here).

Kitaita \ Swahili	AN-SG	AN-PL	3	4	5	6	7	8	9/10	11	15	**Total**
`u`	22	0	0	0	0	0	0	0	2	9	0	**33**
`wi`	0	19	0	0	0	0	0	0	0	0	0	**19**
`ghu`	0	0	16	1	0	0	1	0	2	7	0	**27**
`ghi`	1	0	0	2	0	17	0	2	3	1	0	**26**
`ji`	1	0	0	0	9	1	0	0	2	1	0	**14**
`i`	6	0	2	9	3	3	1	0	11	1	0	**36**
`i_nasal`	4	0	1	1	1	0	0	0	1	0	0	**8**
`chi`	1	0	0	0	1	0	12	0	3	0	0	**17**
`ku`	0	0	2	0	1	0	0	0	2	1	16	**22**
`lu`	0	0	0	0	0	0	0	0	2	2	0	**4**
`ri`	1	0	0	1	2	0	0	1	0	1	0	**6**
`va`	0	1	0	1	0	0	1	14	0	0	0	**17**
Total	36	20	21	15	17	21	15	17	28	23	16	**229**

largest class in this sample, 36 nouns – spans *eight* of the eleven Swahili classes observed, with its largest single Swahili group (9/10) accounting for only 11 of 36 nouns (31%); `ghu`, `ghi`, `ji`, `ku`, and `ri` all spread across five or more Swahili classes as well, each with a plurality rather than a majority in its single most common one. Second, Swahili's own animacy override does not carry over into a single, uniform Kitaita class: the 36 Swahili ANIMATE-SG nouns in this sample resolve to *seven* different Kitaita classes (`u`: 22; `i`: 6; `i_nasal`: 4; `ghi`, `ji`, `chi`, `ri`: 1 each) – a majority take Kitaita's own default animate class `u`, but more than a third (14/36) do not. Separately, the `i`/`i_nasal` split itself (one shared subject marker but two different adjective-concord paradigms, mirroring Swahili's own class 4 vs. 9/10 syncretism) was found only through direct native-speaker correction, after an earlier version of this project's own class table had collapsed the two.

3.4 Substitution, reinflection, and reassembly

Once structure has been identified (§3.2) and each agreement-bearing position's controlling noun has been resolved to a Kitaita class (§3.3), the pipeline's final stage decides, independently for every token at its own original position, what to do with it. Figure 1 summarizes the decision: a token is either a *content* word (noun or verb root), looked up directly in the bilingual dictionary; or an *agreement-bearing slot* identified by the structural stage (subject marker, adjective, possessive, associative, demonstrative, or *-ote* quantifier), reinflected by looking up its controlling noun's Kitaita class in the relevant concord table (Appendix A catalogs every such table); or neither, in which case the original Swahili token is passed through unchanged and flagged as such. Every path writes its result back to the token's own input position – reassembly is purely a *marker-only rewrite*: it never reorders words, inserts new ones, or deletes any (deleting an unrecognized object marker, discussed below, is the one deliberate, documented exception). This is a direct consequence of this project's own validation discipline (§4): a rule that only ever touches a position it has independent evidence for is trivially easier to audit, and to retract, than one that restructures the sentence to fit a guess.

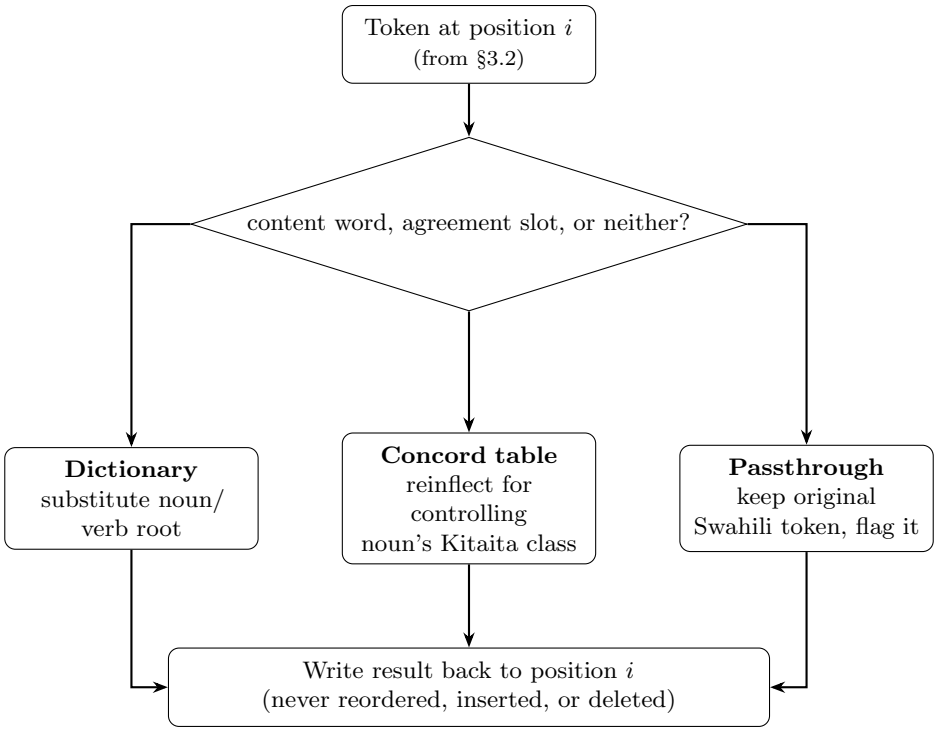


Figure 1: Overview of the substitution/reinflection/reassembly stage, applied independently to every token.

Worked example. *Simba alipanda mti mrefu* ("the lion climbed a tall tree") decomposes into subject *simba* (lion), verb *alipanda* (climbed, past), object *mti* (tree), and adjective *mrefu* (tall) modifying *mti*. The dictionary substitutes *simba*→*shimba* and *mti*→*mudi*; *mudi*'s class resolves to `ghu` (the Swahili class-3 → Kitaita `ghu` mapping is this sample's single largest cell in Table 2, 16/21), so the adjective slot is reinflected via the `refu` concord table (Appendix A) to *mlacha*; *shimba*'s own class, `i_nasal`, giving subject marker *i-*, which the past-tense table (`-li-`→`-de-`) reinflects to *ide-*; the verb root itself, *panda*→*jok*. Reassembled at each token's own original position: *Simba alipanda mti mrefu* → *Shimba idejoka mudi mlacha*.

4 Validation methodology

4.1 Standing validation rules

Every claim in this paper is checked against a small set of rules applied uniformly, not case-by-case, so that a reader can tell how strongly to trust a given number without re-deriving it. First, every probe reported in §3.2 is scored against a control, and we distinguish the two control designs actually used rather than treating them as equivalent: the labelled-edge and classification probes are checked against a Hewitt and Liang (2019) *random-relabel* control (an identical architecture trained on a fixed-but-arbitrary relabelling of the same inputs), with *selectivity* – real accuracy minus control accuracy – reported alongside raw accuracy throughout Table 1; the structural distance probe has so far only been checked against a weaker untrained-probe baseline, and that limitation is stated in §3.2 rather than left implicit. Second, every vocabulary and concord fact in §3.3 and Appendix A is tagged with one of four evidence tiers – VERIFIED (a native speaker judged this specific form directly, via the Kitaita Word Bank), STATISTICAL (a Dice-coefficient

corpus candidate, unconfirmed), HYPOTHESIS (a single data point or a pattern generalized from one), and RESEARCHED-NOT-WIRED – and these tiers are never collapsed into one another; a statistical candidate is reported as a statistical candidate even where it is the only candidate available. Third, no claim in this paper is drawn from $n < 8$ without saying so, and the stability curve in §4.2 below applies this at the level of individual buckets rather than only to the overall n. Fourth, results that fail re-test are retracted explicitly rather than silently replaced: for example, an earlier working hypothesis that adjective concord only needs to resolve a sentence's *subject* noun was retracted in favor of a more general controlling-noun resolution mechanism once non-subject adjective targets were found to fail under the subject-only assumption, and the apparent *gh/ghu* thirteenth Kitaita class used while building Table 2 was corrected mid-construction, not after publication, once it was recognized as a bare-vs-linking- vowel storage artifact of a single class rather than two classes.

4.2 Native-speaker evaluation protocol

The staged plan's final stage (Stage F) calls for evaluating pipeline output with native Kitaita speakers, comparing it against baselines such as naive dictionary substitution and the deterministic adjacency rewrite. This subsection describes the protocol actually built and deployed for that purpose, and reports its current status honestly: the infrastructure is live and has been validated end-to-end on synthetic data, but as of this writing no real native-speaker ratings have been collected, so no evaluation results are reported in this revision.

Deployment. The pipeline is exposed through a public Hugging Face Space ("Kitaita Translator"), so that evaluation does not depend on coordinating a synchronous annotation session: a visitor enters a Swahili sentence, or draws one at random from a pool of 1,059 real sentences (999 mined from all 40 Swahili-language stories in the African Storybook Initiative's open mirror, plus 60 drawn from the Bible parallel corpus for register diversity – roughly 94% storybook register by design, replacing an earlier pool that was 100% Bible-register and risked testing register mismatch rather than the grammar itself), and chooses between two conditions: **Probes (full pipeline)**, the probe-guided rewrite pipeline this paper describes, and **Heuristic (for comparison)**, a plain adjacency-based decomposer kept as a baseline. This realizes two of the conditions from the staged plan's baseline set (§3, Stage F); naive dictionary substitution, counterfactual re-translation, and steering are not yet exposed as selectable conditions in the live tool, which is a real gap against the original plan, noted here rather than silently narrowed. The output is shown together with two collapsed-by-default accordions, "Flags" (identification/translation caveats on that sentence) and "Evidence trail" (the full per-step trace); collapsing them by default means a rater who does not deliberately expand them sees only the raw output, partially – not fully – addressing the staged plan's preference for naive annotators.

What is collected. Per sentence, a visitor gives a holistic 1–5 rating ("1 = way off" to "5 = correct") and, optionally, a free-text corrected Kitaita translation. Each submission is written as its own timestamped JSON file to a private Hugging Face dataset, rather than appended to one shared file, specifically so that two simultaneous submissions cannot clobber each other and a failed upload cannot corrupt data already logged.

From a holistic correction to three structured metrics. The submission form asks only for a holistic corrected sentence, not a per-slot correction – a heavier annotation UI deliberately avoided at this stage. The three metrics below are nonetheless computed at the slot and word level, by position-aligning a correction against the pipeline's own output. This is sound specifically because

this pipeline's reassembly is position-preserving by construction (§3.4): token i of the output and token i of a same-length correction refer to the same original Swahili slot, which is not true of translation pairs in general. A correction whose token count differs from the output's – because the speaker inserted, deleted, or reordered a word – cannot be aligned this way and is excluded from the slot- and word-level metrics entirely rather than guessed at; it still counts toward the plain rating count and mean rating, which require no alignment. All three metrics are computed by the same `kitaita_eval_metrics.py` module used elsewhere in this project, not a parallel implementation built for the dashboard alone:

- **Agreement accuracy**: every agreement-bearing slot the pipeline predicted (subject marker, adjective, possessive, associative, demonstrative, *-ote*) is classified, relative to the aligned correction, as *correct*, *wrong_rewrite* (a rewrite was produced but differs from the correction), *missed_slot* (the correction implies a rewrite the pipeline did not produce), or *spurious_rewrite* (the pipeline rewrote a position the correction leaves unmarked). Counts are pooled by summing across sentences rather than averaging per-sentence accuracies, so a one-slot sentence cannot outweigh a ten-slot one.
- **Vocabulary choice**: for each content-word substitution (noun or verb root, not a concord morpheme), the fraction of independent native-speaker picks for that position that match the pipeline's own word, meaned across scored positions. Positions with zero responses are excluded rather than scored as 0 – absence of data is not the same as disagreement.
- **Stability**: agreement accuracy binned by the exact number of agreement slots a sentence requires, rather than by a pre-chosen range, since gold data at this stage is too scarce for a range boundary to mean anything; any bucket with fewer than 8 sentences is reported but explicitly flagged as not a confident estimate, per the $n \geq 8$ rule in §4.1.

A "Progress" tab on the Space renders all three as charts, refreshed together from one consistent snapshot of the feedback dataset every 60 seconds (plus a manual refresh button and an initial load), so the three charts can never show inconsistent totals relative to each other.

5 Current state and validated findings

6 Open questions and limitations

7 Conclusion and next steps

A Implemented grammatical features

This appendix catalogs every grammatical mechanism the pipeline implements, independent of which example happens to exercise it in the main text. Each item's evidence tier is stated explicitly, in keeping with this project's standing rule (§4): VERIFIED (a native speaker judged this specific form correct, directly, via the project's own *Kitaita Word Bank* data-collection tool); STATISTICAL (a corpus-mined Dice- coefficient candidate from the parallel Swahili–Kitaita Bible corpus, not independently confirmed); HYPOTHESIS (a single data point, or a pattern generalized from one, explicitly flagged in the code as unconfirmed); and RESEARCHED-NOT-WIRED (verified or hypothesized, but not actually called anywhere in the production pipeline). A feature's absence from the final subsection (§A.7) does not mean it is unimplemented – it means it has not yet hit a known failure case worth recording there.

A.1 Nominal morphology

- **Noun class resolution**: three-tier lookup – VERIFIED direct table (subject-marker-per-noun, §3.3) > HYPOTHESIS (deverbal/ class-15-style nouns, borrowed from a sibling research track, never independently tested) > corpus-propagated class (from cross- version agreement in the parallel corpus). A fourth, lowest-confidence tier predicts an UNCLASSIFIED noun's class from its Kitaita word's own 2–3-letter prefix (e.g. `wu-` → `ghu`, 78% majority, $n = 41$ already-classified nouns sharing that prefix) when nothing else is available, and is explicitly flagged as a word-shape HYPOTHESIS, never presented as resolved.
- **Number-invariant nouns** (a fixed set of Swahili nouns that do not change form between singular and plural, e.g. *samaki*/fish): a dedicated singular/plural default subject marker pair, since the ordinary per-noun class lookup cannot distinguish the two readings from the noun alone.
- **Proper names**: an unrecognized, capitalized, non-Bantu-prefix-shaped token defaults to class `u` (mundu's class) for verb agreement – HYPOTHESIS, grounded in the reference-grammar fact that Swahili itself requires any human name to take class-1 agreement regardless of which name it is, not independently re-verified per name.
- **Locative suffix `-nyi`** (mirrors Swahili *-ni*): HYPOTHESIS, a single confirmed example.

A.2 Verbal morphology: subject marking and TAM

Table 3 lists every tense/aspect/mood category the pipeline reinflects. Subject marking itself is resolved two ways, deliberately kept separate: a *class-based* marker when the subject is a real or defaulted noun, and a *person-based* marker (1st/2nd person singular/plural) for a pro-dropped pronoun subject – for a personal pronoun specifically, the VERB's own marker uses the person-based table while any ADJECTIVE modifying it still uses mundu's class `u`, a deliberate decoupling confirmed by direct native-speaker correction (an earlier version of this pipeline wrongly applied the class-based marker to both).

Table 3: TAM categories reinflected by the pipeline.

Swahili category	Kitaita realization	Evidence
Present (*-na-*)	zero-marked (corpus-dominant) or *-we-* (project's stylistic default, also grammatical) + a per-class vowel shift (e.g. `u`→*wa-*)	STATISTICAL, corpus-dominant form confirmed by direct token counts
Past (*-li-*)	*-de-*/*-re-* by class, via a per-class lookup table	VERIFIED, per-class table
Future (*-ta-*)	bare root-initial `k` (most classes) or the fuller root `kai` (classes `wi`/`chi`, plus per-noun exceptions in classes `i`/`ghi`)	VERIFIED, 7/8 nouns confirmed; exceptions tracked per-noun
Narrative (*-ka-*)	*-ka-*, unchanged	VERIFIED
Perfect (*-me-*)	*-da-*	STATISTICAL

Swahili category	Kitaita realization	Evidence
Conditional / participial (*-ki-*)	*-ki-* (hypothesis) or *-ka-* (the project's own independently-attested "conditional" category, per Kitaita's own TAM-allomorphy data)	HYPOTHESIS – which of the two Kitaita variants *-ki-* corresponds to remains unresolved
Copula (*ni*, invariant)	*ni* (no adjective complement) or *dek-*/*kogh-* by class (adjective or existential complement)	VERIFIED
Kuwa na ("to have"), singular	bare root *n* + person/class subject marker, irregular final vowel under negation	VERIFIED, native-speaker-submitted
Kuwa na ("to have"), plural	*[class]+ko* (the SAME existential root as *yuko* below) + comitative *na* before the possessed noun, e.g. *wiko na kazi* ("they have work")	STATISTICAL, reverse-inferred from the Bible corpus, $n \geq 2$ independent verses per form; only the affirmative is wired, negative plural *have* showed inconsistent corpus evidence and was deliberately left unimplemented
Existential/locative (*yuko*/*yupo*/*yumo*)	invariant *oko*	STATISTICAL, $n \geq 15$ corpus verses, zero exceptions – but every tested subject happens to be Swahili class 1/ANIMATE-SG, so invariance across OTHER classes is unconfirmed, flagged explicitly in the code
Passive voice	*-w-* (consonant-final root) / *-lw-* (vowel-final root) infix, mirroring Swahili's own consonant/vowel split	STATISTICAL, 2 independently-confirmed roots
Bare infinitive & infinitive complements (*ku-*)	*ku-* + root + final vowel, kept as a literal dependent infinitive sharing its matrix verb's subject	VERIFIED construction; see §A.7 for the *ku-*/TAM segmentation ambiguity this required resolving

A.3 Negation

Four tenses (present, past, future, subjunctive/imperative), each with a SEPARATE person-based and class-based template – confirmed directly that these are not simple variations of one rule: past uses a dedicated negative-only TAM (*-re-…-eghe*) rather than each class's own affirmative past marker, and an explicit ANIMATE-class exception (classes `u`/`wi`) follows the PERSON template rather than the ordinary noun-class one. All four tenses are VERIFIED, native-speaker-submitted via the Word Bank (2026-10-21), one cell at a time. Negative subjunctive/imperative detection on the Swahili side (*usiogope* vs. *usikilize*, a genuine *-si-* string-level ambiguity) is resolved by comparing root-plausibility evidence for the stripped-vs-unstripped reading, never by string shape alone. The irregular *kuwa na* ("to have") verb keeps its affirmative `-a` final vowel even when negated (*sina*/*huna*/*hana*), mirroring Swahili's own irregularity here rather than following the regular negative templates.

A.4 Agreement/concord rewriting

Adjective concord is the most heavily lexified mechanism: each root has its own confirmed paradigm rather than one shared formula, summarized in Table 4.

Table 4: Adjective concord roots implemented.

Swahili root	Kitaita root/stem	Paradigm shape	Evidence
-baya (bad)	*-zamie*	bare subject-marker prefix, uniform across every class, no mi-/nd- -style split	VERIFIED
-kubwa (big)	*mbaa* default	own irregular/reduced series for classes `ji`/`chi`/`va` (*ibaa*/*kibaa*/*vibaa*); *mbaa* elsewhere	VERIFIED per confirmed class
-zuri (good/nice)	*-ghokie*	class-prefix + shared stem formula	STATISTICAL, Dice 0.40–0.58
safi (clean)	*-elie*	class-prefix + shared stem	HYPOTHESIS
-moja (one)	*-mweri*	class-prefix + shared stem, same formula as *-ote* below	VERIFIED
-ingi (many)	own irregular surface-form set (e.g. *wengi* / *vingi* / *nyingi*), not a class-prefix+stem formula	VERIFIED per confirmed class, *wengi* default otherwise	
-refu (long/tall/deep)	*-lacha*	class `wi`: *walacha* (high confidence, Dice 0.86, 3 versions); all other classes: *mlacha* default (low confidence, Dice 0.53, 1 version)	mixed VERIFIED/STATISTICAL by class
-dogo (small)	*-tini* (default *mtini*)	per-class table	STATISTICAL
-geni (strange/new)	default *mghenyi*	per-class table	STATISTICAL
-fupi (short)	default *mvui*	per-class table	STATISTICAL
-zito (heavy)	default *ulemere*	per-class table	STATISTICAL
-chafu (dirty)	default *ulue*	per-class table	STATISTICAL
-kuteketezwa (burned)	class `va` guess, stem *-kora*	single data point	HYPOTHESIS, $n = 1$
(9 further flat per-(stem, noun) pairs)	–	exact lookup, no class-generalization attempted	VERIFIED

Swahili root	Kitaita root/stem	Paradigm shape	Evidence
(any other adjective root)	default base prefix `m-` (`ji`: `i-`; `chi`: `ki-`; `va`: `vi-`) + the Swahili stem itself, unchanged	HYPOTHESIS, a last-resort generalized formula, flagged as a "double guess" in every output it produces	

Possessive concord: a 12 × 6 grid (every Kitaita class × 1sg/2sg/3sg/1pl/2pl/3pl), VERIFIED, native-speaker-submitted via the Word Bank (60 cells). Singular forms follow a regular i-deletes/u-glides pattern from the class marker; plural *-etu/-enu* show a genuine, reported-not-smoothed- over irregularity (bare-`u`/`ku`/`lu` classes take an *-o-* vowel, every other class an *-e-*).
Associative/genitive concord (*ya/wa/ cha*, ..., "X of Y"): directly VERIFIED for 2 classes (`ghi`→*gha*, `lu`→*lwa*, the latter NOT predictable from the general present-tense vowel-shift pattern the rest of this mechanism reuses); every other class's form is STATISTICAL, extrapolated from that same present-tense vowel-shift table. A separate, descriptive/purposive variant (*-a ku-*VERB, e.g. *maji ya kunywa*, "water for drinking") has its own VERIFIED entry (class `ghi` only) plus the same statistical extrapolation elsewhere.
Demonstrative concord ("this"/"that(near)"/ "that(far)", 3 degrees): full 12-class VERIFIED coverage via the Word Bank, including the `i/i_nasal` split noted in §3.3.
***-Ote* ("all/whole") concord**: class-prefix + shared stem *-ose* formula, VERIFIED for 8 of 12 classes.
Relative clauses (*acl:relcl*): the concord-SOURCE identification probe is validated (Table 1, deployment-condition accuracy 0.836, §3.2), but the Kitaita-side REWRITE of a relative clause's own verb morphology is not robustly wired into the free-text pipeline – confirmed directly, not assumed: *mto ulipofurika* ("the river that flooded") is currently decomposed as two independent clauses rather than one subordinate one, discussed further in §A.7.
Object markers: presence-detection only. The pipeline enumerates candidate (object-marker, alternate-root) readings to correct ROOT identification when a Swahili verb's object marker would otherwise corrupt the statistical root lookup (e.g. *kulifuatilia* = *ku-li-fuatilia*, class-5 object marker *-li-* + root *fuatilia*, not past tense + root *lifuatilia*), but the object marker itself is a deliberate **scoping decision**, not a gap: it is dropped rather than reinflected into Kitaita object agreement ("marker-only rewrite rule: object agreement optional, matches Swahili source"). A separate hypothesis table for person-based Kitaita object markers (1sg *ni*, 1pl *di* – notably *different* from Swahili's own *tu* – and a combined class-2/2pl *wi*, each HYPOTHESIS-tier, n 4–10) exists from dedicated research but is RESEARCHED-NOT-WIRED: not called anywhere in the production pipeline.

A.5 Idiomatic and periphrastic verb constructions

Four Swahili verbs do not decompose as a single root at all, and are each handled by their own confirmed periphrasis rather than forced into the regular root+TAM template:

- *tawala* (rule) = *bony* (do/make) + fixed object *nguma* (authority) – STATISTICAL, confirmed compositionally (both halves independently attested elsewhere in this project's data, not just as one fixed phrase).
- *kasirisha* (anger, causative) = *josh* (raise) + [object] + fixed object *machu* (anger) – VERIFIED, direct user correction.

- *fifia* (fade/grow cold) = *hor* + fixed object *modo* (cold) – HYPOTHESIS, $n = 1$, a Word Bank predicate for "love" specifically.

- *furahi*/*kasirika* (enjoy; get angry) – a matched passive/causative pair: both are INHERENTLY PASSIVE in Kitaita with no active transitive form at all in the corpus, taking a *ni*-agent phrase ("pleased/angered BY X") rather than a plain direct object (*furahi*: *boi* → passive *boilw*-, STATISTICAL, $n = 620$ corpus tokens, zero active-transitive counterexamples; *kasirik*: *jok* → passive *jokw*- + fixed agent *ni machu*, STATISTICAL, $n = 8$ corpus verses, zero exceptions, the passive counterpart of *kasirisha* above – one unified finding, not two unrelated ones).

A.6 Vocabulary substitution

Noun and verb-root translation is itself tiered: a VERIFIED direct lookup first; then a STATISTICAL corpus-mined candidate (Dice-coefficient co-occurrence, cross-checked for agreement across the 3 independently-translated Swahili Bible versions in the parallel corpus); then a weaker per-surface-form statistical fallback when no root-level candidate exists at all (flagged as such, since it cannot be safely reinflected – its own internal morpheme boundaries are unknown); and finally, when no Kitaita vocabulary exists for a root at any tier, a "Taita-fied hybrid" that keeps the original Swahili root but applies Kitaita subject/TAM morphology around it – explicitly presented as a readability stopgap, never as a real translation.

A.7 Fallback and default strategies

Used only when no noun-specific evidence exists at all, and always flagged as an unconfirmed guess in the output rather than presented as resolved:

- A noun with no Kitaita translation at all still gets VERB agreement from a majority-default marker for its Swahili class (ANIMATE-SG→`u`, ANIMATE-PL→ `wi`, and 7 further shape-classes), with a short, explicit exclusion list (*mbwa*/*nyoka*/*simba*/ *ndege*/*chui*/*samaki*) of animate nouns known NOT to follow this default.

- A 3rd-person pro-dropped subject with no antecedent in the sentence defaults to class `u`/`wi` (mundu/wandu) directly from Swahili's own *a*-/*wa*- marker meaning – an unconfirmed guess about *which* noun, not about which class, and explicitly out of scope as a coreference-resolution mechanism.

- An unrecognized token with a Bantu noun-class-prefix SHAPE (e.g. *ki*-/*vi*-) but zero vocabulary coverage can still supply a class guess for agreement purposes from its own prefix shape alone.

Known, currently unresolved gaps, stated plainly rather than hidden: coordinated-subject agreement (which plural form – closest-conjunct, shared-class, or human-plural-override – governs a coordinated subject has no settled rule, and three different sources – the gold corpus, a naive coordination rule, and NLLB's own generation – gave three different answers for one real test case); relative-clause rewrite, as noted above; and the *ku*-/TAM segmentation ambiguity (Swahili *ku*- is genuinely ambiguous between the infinitive marker, which never takes a further tense marker, and class-17/ locative subject agreement, which does – e.g. *kutakuwa*, "it will be" – resolved by enumerating both readings and defaulting to the infinitive one specifically at the two points in the pipeline that already commit to that reading, not by picking one globally).

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